

## LOGARITHM

- **Logarithm**: Logarithm is the opposite of powers.  
If  $a = b^c$  then its logarithmic form is  $\log_b a = c$  (log of a to the base b is c).  
It is defined only for  $a > 0$ .
- **Common Logarithm**: Logarithm with base 10 is called as common logarithm. It is denoted by  $\log$ .
- **Natural Logarithm**: Logarithm with base e is called as natural logarithm. It is denoted by  $\ln$ .
- $\log_a(mn) = \log_a(m) + \log_a(n)$ .
- $\log_a(m/n) = \log_a(m) - \log_a(n)$ .
- $\log_a(m^n) = n\log_a(m)$ .
- $\log_a(1) = 0$
- $\log_a(1/x) = -\log_a(x) = \log_{\frac{1}{a}}(x)$ .
- $x = e^{\ln x}$ .
- $\log_a(x) = \frac{\log_b(x)}{\log_b(a)}$ .
- $\log_a(x) = \frac{1}{\log_x(a)}$ .
- $\log_a(a) = 1$ .
- $a^{\log_a(x)} = x$ .
- $\log 10 = 1$ .
- $\log 10^n = n$ .
- $y^{\log_a(x)} = x^{\log_a(y)}$ .
- $\log_{a^n}(x) = \frac{\log_a x}{n}$ .
- $\log_{a^n}(x^n) = \log_a(x)$ .

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